AGRICULTURAL DRAINAGE · NOTE DR03

Surface Drainage Design — Drain Types, Curve Number Runoff, Manning Sizing and Field Ditch Systems

Course topics: CAE1.3 L14, IDE-502 T6, IDE-502 T7, IDE-502 T8

What drains do

Drains lower the groundwater and keep it below the crop root zone, and they carry rainfall runoff away quickly. Open ditches remove the excess runoff; deep open ditches can also act as subsurface drains by lowering the water table below the root zone. Water enters a ditch directly from the surface or by seepage from the surrounding soil.

Drain type	Where and how it works
Interceptor drain (surface or subsurface)	Runs along hills, canals, dikes or reservoirs, usually **perpendicular** to the groundwater flow. It intercepts seepage and runoff and leads it away, isolating the problem area from the source of excess water. It is often placed along field boundaries to interfere less with farming; the ideal place is along the foothills where seepage enters flat land
Relief drain (open, subsurface or vertical)	Relieves saturated soil of excess water in the subsoil or at the surface. It runs **parallel** to the groundwater flow and water seeps in from both sides

Planning a field drainage system

1. Diagnose the drainage problem correctly.
2. Investigate and survey for physical and economic feasibility, with a clear picture of the land, water resources and crop needs.
3. Begin with a **reconnaissance survey** to decide feasibility, the type of system and its approximate cost.
4. Plan the four components: outlet, main drains, field drains and auxiliary structures.

Principal method	What it is
Surface drainage	A network of open channels plus land grading
Subsurface (pipe) drainage	Pipes of clay tiles, concrete, plastic or perforated conduit. Water enters through joints, slots or perforations, flows to a collector and on to the main drain. Pipe drains intercept lateral groundwater flow, lower the water table and remove excess root-zone moisture
Mole drains	Unlined cylindrical channels formed by a mole plough — hydraulically like pipe drains but temporary (effective only for a few years)
Vertical drainage	Pumping groundwater from many tubewells across the area lowers the water table and reduces salinity; the pumped water can supplement irrigation. The well must be connected to the water table
Bio-drainage	Quick-growing trees and plants that use a lot of soil water — eucalyptus, casuarina, *Bambusa arundinacea*, tamarix and lucerne

Preventing waterlogging and salinity: line the irrigation distribution system, manage water efficiently on the farm, and maintain surface drains with farmer participation.

Surface drainage

Surface drainage removes excess water from the land surface — water from rainfall, over-irrigation, conveyance losses or seepage.

- **Suits:** flat or level land with impermeable subsoil; places where pipe drainage is impractical or uneconomical; removing large storm flows.
- **Favourable conditions:** very slowly permeable soils (heavy clay); shallow soils (20–50 cm) over slowly permeable subsoil; no suitable outlet for tile drains; high-intensity rainfall (monsoon areas); humid regions that need rainwater removed fast.
- **Drawbacks of open drains:** they cut up the farm and hinder machinery, need frequent cleaning and harbour weeds.
- **Components:** on-farm field drains (relief drains), intermediate or collector drains, main or sub-main drains, interceptor drains, and land grading and smoothing.
- **Construction rule:** start at the outlet and work upstream.

Land smoothing

- Shapes the land into a plane surface with a uniform slope that helps drainage: cut the high spots and fill minor depressions so water runs off naturally — a smooth surface, not perfect levelling.
- Control the depth of grading so unproductive subsoil is not exposed; leave the surface smooth and free of depressions; put field drains where depressions are too deep to fill.
- Work in two stages: rough grading, then final smoothing with animal-drawn or tractor-drawn equipment.

Open drains

- **Field drains** remove surface water from fields; **outlet drains** give an outlet to tile drains and tributary channels. Design follows open-channel practice, with capacity increasing downstream; they are needed on low-lying, flat land with little slope.

- An open-ditch system drains a whole watershed: field ditches → outlet ditches → mains, with lateral outlets spaced widely to disturb farming less.
- **First design step:** fix the outlet and its operating water levels. The cheapest outlet is a natural waterway (stream or river). The main drain occupies the lowest ground, starting at the outlet and running upslope along natural depressions.
- **Rate of removal** depends on rainfall, drainage area, runoff characteristics (slope, soil, vegetation) and crop. Remove the excess within 12, 24 or 48 hours after heavy rain; some overflow is acceptable for ordinary crops, but vegetables are highly flood-sensitive.

Design procedure

1. **Hydrologic design** — estimate the runoff, i.e. the drainage coefficient.
2. **Hydraulic design** — size the section with Manning's formula.

A properly designed open drain has enough capacity for the design flow, a water surface low enough to drain the land, stable side slopes, and a velocity that neither scours nor silts.

Estimating runoff: rational and Curve Number methods

Use the **rational method** for small watersheds and the **Curve Number (CN) method** of the US Soil Conservation Service for medium to large basins. CN, from 0 to 100, expresses a watershed's runoff potential; it was derived from rainfall–runoff studies of many US watersheds.

- **Design rainfall:** if the time of concentration is under 6 hr, use the 6-hr rainfall, usually with a 5- or 10-year recurrence interval. Convert 24-hr rainfall to 6-hr rainfall by multiplying by 0.6 (range 0.5–0.7), and distribute it within the 6 hr using the WMO guideline table.

Hydrologic soil group	Infiltration rate (as in the notes)	Runoff potential
A	> 25 mm/hr	Low
B	12.5–25 mm/hr	Moderately low
C	2.5–12.5 mm/hr	Moderately high
D	< 2.5 mm/hr	High

Check: USDA TR-55 defines the groups by minimum infiltration rate of bare soil after prolonged wetting — A > 7.6, B 3.8–7.6, C 1.3–3.8 and D < 1.3 mm/h. Quote the TR-55 values if a question cites the USDA definition.

Antecedent moisture condition	Runoff	Hydrologic condition (canopy)	Runoff
AMC-I — dry	Low	Poor: < 50%	Heavy
AMC-II — average	Medium	Fair: 50–75%	Medium
AMC-III — wet	High	Good: > 75%	Low

- **Initial abstraction** I_a = 0.1S, 0.2S or 0.3S; 0.2S is the commonly accepted value. In India: black soils at AMC II and III use I_a = 0.1S; black soils at AMC-I and all other soils use I_a = 0.3S.

$$\text{Potential retention: } S = (1000/CN - 10) \times 25.4 \text{ mm}$$
$$\text{Direct runoff: } Q = (R - I_a)^2 \,/\, (R - I_a + S) \text{ mm } (R = \text{rainfall, mm})$$

Correction: the notes print the runoff formula as $(R - I_a)^2/(R + (S - I_a)) \times 25.4$ mm. When R and S are already in mm, Q comes out in mm directly — the extra × 25.4 belongs only to the S equation (it converts inches to mm).

Tested on watersheds in India and the USA, the CN method showed an average deviation of about 3.34 mm between estimated and observed runoff, with most cases under 10 mm — so it gives good runoff estimates.

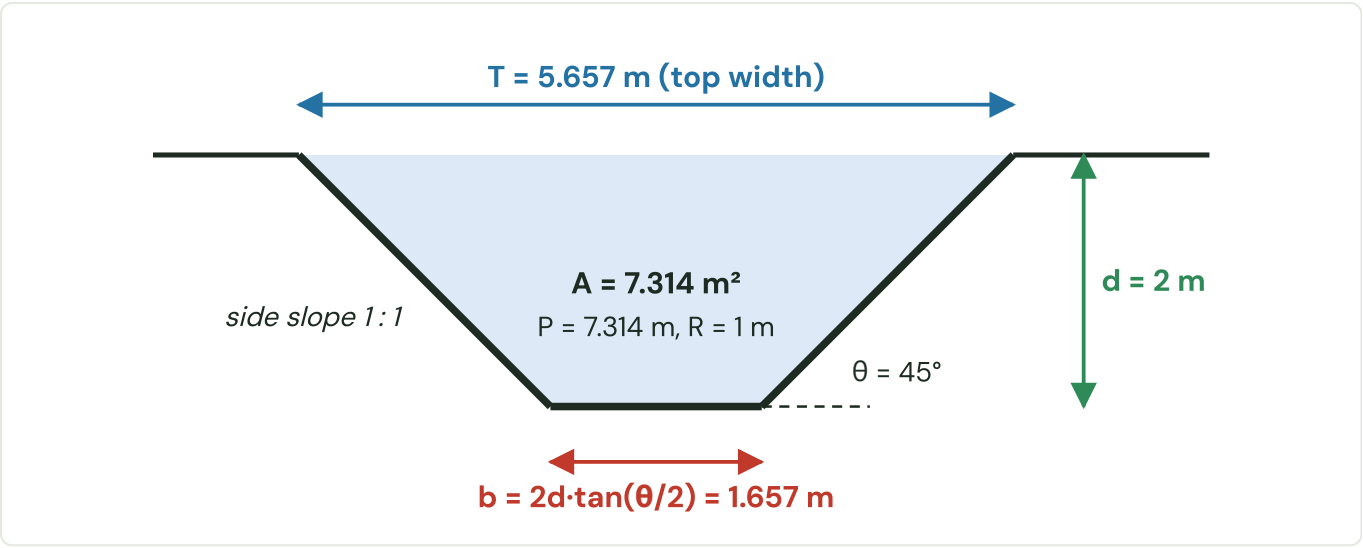
Hydraulic design of open drains

- **Capacity:** the section must carry the drainage flow; predict the design peak with the rational formula or the CN-based drainage coefficient. Where overflow is not wanted, keep the design water surface (hydraulic grade line) about 30 cm below the field surface.
- **Section:** trapezoidal, sized by velocity and quantity using Manning's formula, v = (1/n)·R^(2/3)·S^(1/2).
- **Manning's n:** 0.04 for bottom width ≤ 1.2 m; 0.035 for larger ditches.
- **Depth:** enough to drain the area. Channels that are outlets for tile lines generally need depth ≥ 1.2 m and at least 30 cm clearance between the tile flow line and the normal water level in the channel.
- **Grade:** set by topography and soil; follow the natural ground slope along the ditch, cutting the bed deeper on flat land to get the needed grade. Permissible velocity depends on soil texture.

Soil	Side slope (H : V)	Soil	Maximum velocity (m/s)
Clay	1 : 1	Sand and sandy loam	0.75
Silt loam	1.5 : 1	Silt loam	0.90
Sandy loam	2 : 1	Sandy clay loam	1.00
Loose sandy soil	3 : 1	Clay loam	1.20
		Stiff clay	1.50

Most efficient trapezoidal section (least excavation): $b = 2d \cdot \tan(\theta/2)$
b = bottom width, d = depth, θ = angle of the side slope with the horizontal

- **Spoil and berms:** spread excavated spoil upslope to merge with the natural surface. If it is heaped in banks, leave a clear berm ≥ 3 m between ditch and bank for cleaning, weed removal and inspection — especially by large machines. Seed berms and side slopes against erosion.
- **Alignment:** avoid sharp bends; prefer long, flat curves.
- **Construction equipment:** ploughs, V-ditchers, scrapers, elevating graders, bulldozers, draglines or clamshells, and dynamite where required.



Example 55 — most efficient trapezoidal drain in clay, side slope 1 : 1 (drawn to scale).

Example 55 — hydraulic design of a drainage channel in clay
Given: flow depth d = 2 m; side slope 1 : 1 (θ = 45°); grade 0.04% (S = 0.0004); permissible velocity in clay 1.2 m/s.
1. Bottom width: b = 2 × 2 × tan 22.5° = 1.657 m
2. Top width: T = b + 2zd = 1.657 + 2 × 1 × 2 = 5.657 m (the notes write T = 2d + b, true only for a 1 : 1 slope)
3. Area: A = (b + zd)d = (1.657 + 2) × 2 = 7.314 m²
4. Wetted perimeter: P = b + 2d√(1 + z²) = 1.657 + 5.657 = 7.314 m
5. Hydraulic radius: R = A/P = 1 m
6. Velocity (n = 0.035, since b > 1.2 m): v = (1/0.035) × 1^(2/3) × (0.0004)^(1/2) = 0.57 m/s — below the 1.2 m/s limit, so the bed will not scour
7. Discharge: Q = A·v = 7.314 × 0.57 = 4.17 m³/s
(Recomputed: all seven values check.)

Levees

- Levees are earth embankments built along streams to prevent flooding during high runoff. They add capacity beyond an ordinary drain design by raising the channel's water surface (hydraulic grade line) above the surrounding land and increasing the cross-section and velocity.
- With levees, the ditch drainage coefficient must handle a 10-, 25- or 50-year recurrence storm. Side slopes, bottom width and berm width follow open-ditch design; spoil from the excavation builds the levee.
- **Culverts** pass drainage water through the levee when the stream is low. **Flood gates** open when the water level is higher on the land side than in the ditch, and may be operated by hand.
- Build levees on earth-dam principles, with grass cover against erosion.

Ponded areas

- Low spots collect runoff from higher ground and hold it for days or weeks. Drain them with **random field ditches** or regular field surface drains.
- **Random ditches** are shallow with flat side slopes that machines can cross: 8 : 1 or flatter where operations cross the ditch, at least 4 : 1 when tillage runs parallel to it. Align them for minimum cuts and least interference with farming.

- Other options: a surface inlet to a tile line, a tube outlet, or — if the depression is too deep — pumping to a nearby tile drain or adjoining area.

Flat land (slope < 1.5%)

Use land smoothing plus field ditches — either a bedding system or parallel field ditches.

Bedding system

- Water drains sideways from crowned beds → dead furrows → row drains → collection ditches → outlet. The area between two dead furrows is a **bed**.
- Run dead furrows in the direction of greatest slope; plough parallel to them and plant and cultivate across them. Dead furrows need a continuous, unobstructed grade.
- **Bed width:** 7–11 m on very slowly draining soils, 13–15 m on slowly draining soils. **Bed depth:** 15–45 cm, half of which forms the dead furrow. **Bed length:** 90–300 m.
- **Collector drains** run across the slope at regular intervals — about 90 m apart on nearly flat, very low-permeability soils and about 300 m on more sloping soils of fair permeability.
- Form beds by back-furrowing with a mouldboard plough (several ploughings may be needed); maintain them by always turning the furrows the same way.

Parallel field ditch system

- Like bedding, but the ditches are farther apart and carry more than dead furrows. Field drains usually meet the collector at right angles; where that is impossible they join at an angle (< 90°) — a herringbone layout. Suits flat, poorly drained soils with many shallow depressions.
- **Design features:** minimum ditch depth 22.5 cm; minimum cross-section 0.5 m²; side slopes 8 : 1 or flatter so machines can cross; rows may run at most 180 m in one direction, so ditches may be at most 360 m apart when rows drain both ways.
- Sections may be V-shaped, trapezoidal or parabolic. A **W-ditch** is two parallel ditches with the spoil between them, at least 9 m between centre lines — best for flat land draining from both sides.
- **For groundwater control** the ditches may be deeper and uncrossable: spacing 60 m in clay and 200 m in sandy soil; minimum depth 1.2 m in sand and 0.9 m in clay. Use the spoil to fill depressions or spread it for cultivation.

Sloping land: cross-slope ditch system

- For sloping wet land: ditches run around the slope on a uniform grade, following the natural lie of the land.
- Suits slopes of about 4% or less with poor internal drainage and shallow depressions that hold water — land too steep for bedding or field ditches, where up-and-down farming would cause erosion.
- Lay out like terraces but with little or no ridge on the downhill side: use the excavated soil to fill low areas or spread it downhill so any ridge is ≤ 7.5 cm high, then smooth the land.
- Success depends on eliminating low spots (no ponding) and on tilling parallel to the channels.

Sources: ICAR eCourse 'Drainage Engineering' — Dr M. K. Jha (IIT Kharagpur), reviewed by Prof. K. Yellareddy (Director A&R, WALAMTARI, Hyderabad); AgriMoon e-book. USDA-NRCS TR-55, Urban Hydrology for Small Watersheds (hydrologic soil groups). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. For the most efficient trapezoidal drain section, the bottom width is

(a) b = d·tan θ
(b) b = 2d·tan(θ/2)
(c) b = d/2
(d) b = 2d·cot θ

Q2. Manning's n for an open drain with bottom width ≤ 1.2 m is taken as

(a) 0.013
(b) 0.025
(c) 0.035
(d) 0.04

Q3. A clay drain is 2 m deep with 1 : 1 side slopes. Its most efficient bottom width is about

(a) 1.0 m
(b) 1.657 m
(c) 2.0 m
(d) 4.0 m

Q4. Maximum velocity in an open ditch in stiff clay is about

(a) 0.75 m/s
(b) 0.90 m/s
(c) 1.20 m/s
(d) 1.50 m/s

Q5. In the CN method the commonly accepted initial abstraction is

(a) 0.1S
(b) 0.2S
(c) 0.3S
(d) 0.5S

Q6. Recommended side slope of an open drain in loose sandy soil is

(a) 1 : 1
(b) 1.5 : 1
(c) 2 : 1
(d) 3 : 1

Q7. Random field ditches that machines cross should have side slopes of

(a) 2 : 1
(b) 4 : 1
(c) 8 : 1 or flatter
(d) 1 : 1

Q8. Bed width in a bedding system on very slowly draining soils is

(a) 3–5 m
(b) 7–11 m
(c) 13–15 m
(d) 30–40 m

Q9. The cross-slope ditch system suits slopes of about

(a) 4% or less
(b) 8–12%
(c) 15–20%
(d) over 25%

Q10. An interceptor drain is usually aligned

(a) parallel to groundwater flow
(b) perpendicular to groundwater flow
(c) along the steepest slope
(d) diagonally

Answers

Q1 — (b) b = 2d·tan(θ/2)
Q2 — (d) 0.04
Q3 — (b) 1.657 m — *2 × 2 × tan 22.5°.*
Q4 — (d) 1.50 m/s
Q5 — (b) 0.2S
Q6 — (d) 3 : 1
Q7 — (c) 8 : 1 or flatter
Q8 — (b) 7–11 m
Q9 — (a) 4% or less
Q10 — (b) perpendicular to groundwater flow